

SEMANTIC WEB!

WEB 3.0

INTRODUCTION!

- Web 3.0, also known as the Semantic Web, is the next evolution of the World Wide Web.
- It aims to create a more intelligent, connected, and decentralized web by integrating technologies such as artificial intelligence (AI), blockchain, and linked data.
- **Web 1.0** → Static web pages (read-only)
- **Web 2.0** → Interactive & social web (read-write)
- **Web 3.0 (Semantic Web)** → Intelligent, decentralized, and machine-readable web (read-write-execute)

CHARACTERISTICS!

The Semantic Web is an extension of the current web where data is linked and understood by machines. It introduces structured data that enables automated agents and AI to process, interpret, and derive meaningful insights.

1. MACHINE READABILITY

- The Semantic Web enables machines to understand and process web data instead of just displaying it to humans.
- Uses structured data (RDF, OWL) to provide context and meaning to information.
- AI and Natural Language Processing (NLP) help extract insights from unstructured data.

Use Case:

- Chatbots & Virtual Assistants (e.g., Siri, Alexa) use semantic data to provide relevant responses.
- Healthcare Systems use machine-readable medical records for AI-powered diagnostics.

CHARACTERISTICS!

2. INTEROPERABILITY

- Different systems and applications can share and interpret data in a consistent way.
- Uses standardized frameworks like RDF, OWL, and SPARQL to enable seamless data exchange.
- Helps in cross-platform data integration without compatibility issues.

Use Case:

- Smart Cities integrate traffic, weather, and transport data across multiple platforms.
- E-Governance allows different agencies (e.g., tax, health, law enforcement) to share citizen records securely.

CHARACTERISTICS!

3. DECENTRALIZATION

- Unlike Web 2.0, where data is stored in centralized databases, the Semantic Web supports distributed data sources.
- Uses blockchain and decentralized identifiers (DIDs) for better security and ownership of data.
- Reduces reliance on big tech companies controlling user data.

Use Case:

- Decentralized Finance (DeFi) uses smart contracts to remove banks as intermediaries.
- Self-sovereign identity systems (SSI) allow users to control their personal data instead of companies like Facebook or Google.

CHARACTERISTICS!

4. PERSONALIZATION

- AI and ML analyze semantic data to deliver personalized content.
- The web understands user preferences, search history, and interactions to improve recommendations.
- Enhances context-aware computing for tailored user experiences.

Use Case:

- E-commerce (Amazon, eBay) uses personalized product recommendations.
- Streaming Services (Netflix, Spotify) suggest content based on user behavior.

CHARACTERISTICS!

5. AUTOMATION

- The Semantic Web enables intelligent agents (bots) to perform tasks autonomously.
- Uses ontologies and knowledge graphs to link and interpret complex relationships.
- Reduces manual efforts in data retrieval, analysis, and decision-making.

Use Case:

- Autonomous Vehicles use semantic data for real-time traffic and navigation.
- AI-Powered Resume Screening automates candidate selection based on job descriptions.

CHARACTERISTICS!

6. DATA INTEGRATION & LINKED DATA

- The Semantic Web connects related data across different sources.
- Uses Linked Data principles to ensure datasets are reusable and queryable.
- Enables cross-domain knowledge representation.

Use Case:

- Google Knowledge Graph connects entities like people, places, and events.
- Medical Research integrates patient records, genetic data, and clinical trials.

COMPONENTS!

1. RDF (Resource Description Framework)

- A framework for representing information about resources on the web in a structured way.
- Uses triples: Subject → Predicate → Object
- Example:

```
<rdf:RDF>
  <rdf:Description rdf:about="http://example.com/person1">
    <name>John Doe</name>
    <age>30</age>
    <occupation>Engineer</occupation>
  </rdf:Description>
</rdf:RDF>
```

COMPONENTS!

2. OWL (Web Ontology Language)

- Defines relationships between concepts in a way that computers can understand.
- Example:

```
<owl:Class rdf:ID="Person"/>
<owl:Class rdf:ID="Engineer"/>
<owl:ObjectProperty rdf:ID="hasOccupation">
  <rdfs:domain rdf:resource="#Person"/>
  <rdfs:range rdf:resource="#Engineer"/>
</owl:ObjectProperty>
```

COMPONENTS!

3. SPARQL (Semantic Query Language)

- A query language for retrieving data from RDF.
- Example:

```
SELECT ?name ?occupation
WHERE {
    ?person <http://xmlns.com/foaf/0.1/name> ?name.
    ?person <http://xmlns.com/foaf/0.1/occupation> ?occupation.
}
```

COMPONENTS!

4. Linked Data

- Data is linked across different websites, allowing for data integration across multiple sources.
- Example:

```
@prefix dbpedia: <http://dbpedia.org/resource/> .  
@prefix foaf: <http://xmlns.com/foaf/0.1/> .  
  
dbpedia:Albert_Einstein foaf:name "Albert Einstein" .
```

5. Reasoners & AI Integration

- AI-powered reasoning systems interpret ontologies and make logical inferences.

WEB STACK!

The Semantic Web Stack (also called the Semantic Web Layer Cake) consists of several interconnected layers:

1. Data Representation Layer (Base Layer)

- URI & IRI (Internationalized Resource Identifier) – Identifies web resources uniquely.
- XML (Extensible Markup Language) – Structures data for easy parsing.
- RDF (Resource Description Framework) – Provides a standardized way of describing relationships between resources.

2. Ontology Layer

- RDFS (RDF Schema) – Provides a basic vocabulary for defining data relationships.
- OWL (Web Ontology Language) – Defines richer, more complex relationships between data.

WEB STACK!

3. Query & Inference Layer

- SPARQL – Used to query RDF-based data sources.
- Inference & Logic – Enables reasoning over data to infer new relationships.

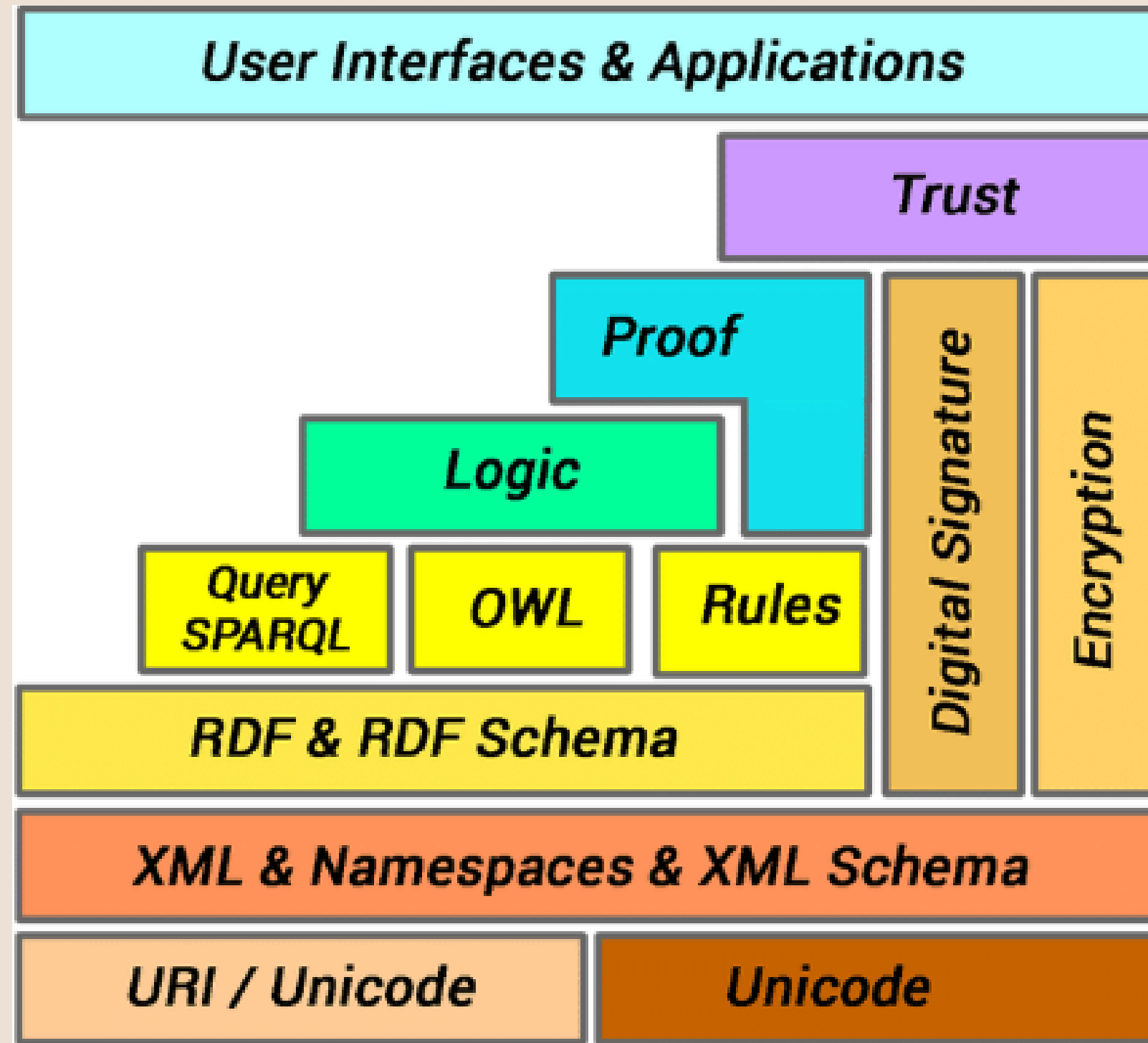
4. Trust and Proof Layer

- Cryptography & Digital Signatures – Ensures security, trust, and authenticity of data.
- Blockchain (Web 3.0) – Helps decentralize data and ensure tamper-proof transactions.

5. User Interface & Applications Layer

- Web applications and intelligent agents that leverage the underlying structured data.

WEB STACK!



THANK YOU!